

Current Calculation

Experiment 3 (Series)

$$V = 5V, R_1 = 1.034 \text{ k}\Omega, R_2 = 1.044 \text{ k}\Omega$$

$$I = ?$$

$$R_T = R_1 + R_2$$

$$R_T = 1.034 \text{ k}\Omega + 1.044 \text{ k}\Omega$$

$$= 2.078 \text{ k}\Omega$$

$$= 2.078 \times 1000$$

$$= 2078 \Omega$$

$$V = IR, I = \frac{V}{R}$$

$$\therefore \frac{5}{2078}$$

$$\text{Ans} = 0.002406 \text{ A}$$

Experiment 4 (Parallel)

$$V = 5V, R_1 = 1.914 \text{ k}\Omega, R_2 = 1.714 \text{ k}\Omega$$

$$I = ?$$

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$1.914 \times 1000 = 1914 \Omega$$

$$1.714 \times 1000 = 1714 \Omega$$

$$\frac{1}{1914} + \frac{1}{1714} = \frac{907}{820149}$$

$$R_T = \frac{820149}{907} = 907.24 \, \Omega$$

$$V = IR, \quad I = \frac{V}{R}$$

$$\therefore \frac{5}{907.24}$$

$$Ans = 0.00553 \, A$$